Geological Remote Sensing Interpretation Using Deep Learning Feature and an Adaptive Multisource Data Fusion Network

Wei Han, Jun Li, *Fellow, IEEE*, Sheng Wang, Xinyu Zhang, Yusen Dong, Runyu Fan, Xiaohan Zhang, and Lizhe Wang, *Fellow, IEEE*

***Abstract*— Geological remote sensing interpretation can extract elements of interest from multiple types of images, which is vital in geological survey and mapping, especially in inaccessible regions. However, due to numerous classes, high interclass similarities, complex distributions, and sample imbalances of geological elements, the interpretation results of machine learning (ML)-based methods are understandably worse than manual visual interpretation. In addition, scholars in remote sensing have mainly carried out their works to interpret a single geological element category, such as mineral, lithological, soil, and structure. The interpretation of multiple geological elements is missing, which is more in line with the open world. To improve the interpretation results of ML-based methods and reduce the labor cost in geological survey and mapping, we propose a deep learning (DL)-feature-based adaptive multisource data fusion network (AMSDFNet) for the efficient interpretation of multiple geological remote sensing elements. The AMSDFNet has two branches for learning valuable spatial and spectral information from two kinds of data sources, in which the atrous spatial pyramid pooling (ASPP) operation and an attention block are applied to adaptively extract and fuse multiscale informative features. A hard example mining algorithm was also added to select important training examples to address sample imbalance. A large-scale region in western China with sufficient geological elements was set as the research area. The proposed model improved the two critical metrics by about 2% in the experiment section. As far as we know, this research work is the first time DL features and multisource remote sensing images have been utilized to simultaneously interpret geological elements of lithology, soil, surface water, and glaciers. The extensive experimental results demonstrated the superiority of DL features and our model in geological remote sensing interpretation.**

***Index Terms*— Deep learning, geological remote sensing, multisource data fusion, semantic segmentation.**

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Wei Han, Jun Li, Sheng Wang, Xinyu Zhang, Yusen Dong, Runyu Fan, and Xiaohan Zhang are with the School of Computer Science and the Hubei Key Laboratory of Intelligent Geo-Information Processing, China University of Geosciences, Wuhan 430078, China (e-mail: weihan@cug.edu.cn).

Lizhe Wang is with the School of Computer Science, the Hubei Key Laboratory of Intelligent Geo-Information Processing, and the Key Laboratory of Geological Survey and Evaluation of the Ministry of Education, China University of Geosciences, Wuhan 430074, China (e-mail: lizhe.wang@gmail.com).

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I. Introduction

GEOLOGICAL remote sensing interpretation applies advanced data processing and information extraction technologies to obtain elements of interest from image taken from sources such as satellites and aircraft and is of great significance for research in analyzing the process and characteristics of geological evolution. Typical remote sensing geological elements include minerals, lithology, structures, soil, surface water, glaciers, various disasters, and so on [1].

In the past decades, as the spatial, spectral, and temporal resolutions of Earth observation technologies have been continuously improved [2], [3], geological remote sensing interpretation has increasingly played a critical role in geological survey and mapping, especially in inaccessible regions. Many research scholars have started to apply machine learning (ML) methods with various types of images in geological remote sensing interpretation, in particular for mineral [4], lithological [5], glacier [6], soil [7], and structural mapping [8]. These valuable works have significantly promoted the development of geological remote sensing. However, due to the many categories of geological elements with similar chemical compositions, high interclass similarities, complex spatial distributions, and unbalanced samples, there is still a big gap between the accuracy of ML-based remote sensing geological interpretation and manual visual interpretation. The factors affecting the interpretation of geological remote sensing can be summarized in the following.

1) *Numerous Classes of Geological Elements:* A remote sensing image always covers an extensive region of the Earth's surface and contains many types of geological elements, including rocks, soil, water, and glaciers. Each of these categories also has multiple subcategories. Numerous geological element classes necessitate higher requirements for the interpretation models.
2) *High Interclass Similarity:* Different geological elements have a similar chemical composition, color, shape, and texture feature, which makes it difficult to identify them correctly. Taking lithology as an example, slate and phyllite have very similar chemical compositions, and their main differences are in structure and shape. However, the differences between them are further reduced in remote sensing images. These elements with similar

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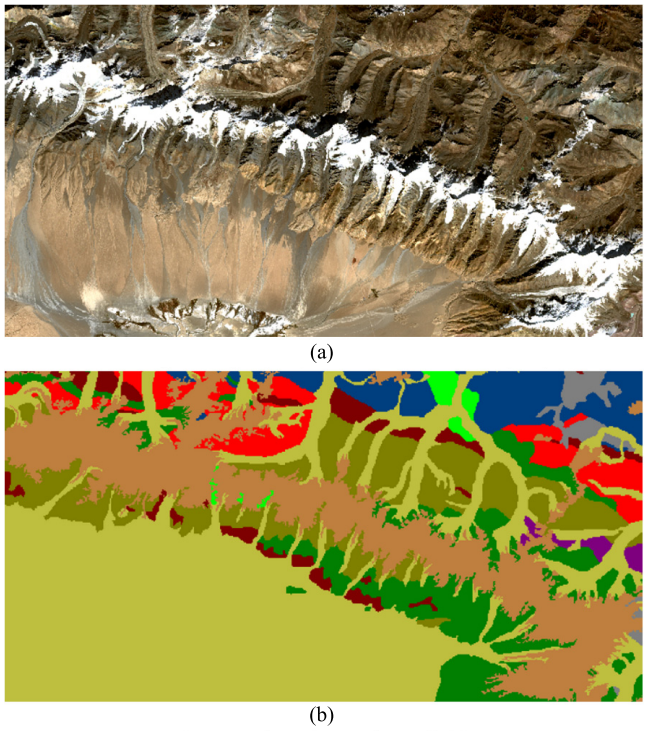


Fig. 1. Example of the geological remote sensing images. (a) Sentinel 2 image (example of the geological environment remote sensing images) and (b) annotation map (corresponding label map of image). (The category information represented by each color can be seen in Fig. 3.)

chemical compositions lead to serious homogeneity. In addition, due to the influence of geological movement, weathering, vegetation, and imaging conditions, some elements' visual features and category information are apt to change or form a mixed region with blurred boundaries.

3) *Complex Distribution and Sample Imbalance:* In Fig. 1, the distributions of different geological elements are extremely complex, and the sample numbers of the elements also vary greatly. A small number of geological element categories occupy a large proportion of the Earth's surface and thus also the main part of the sample pixels. By contrast, many other geological element types are concentrated in small regions and occupy a small number of pixels. This severe sample imbalance usually affects the training process of the interpretation model, resulting in limited interpretation accuracy.

The above factors make geological remote sensing interpretation a challenging task. To efficiently interpret geological elements from remote sensing images, many domain scholars have presented different models to improve the interpretation performance. Because each category of geological elements has a diagnostic spectral response that can be recorded by multispectral and hyperspectral remote sensing images, scholars initially used spectral matching and analysis algorithms to classify different geological elements. Representative methods are band ratioing [9], spectral angle mapper (SAM) [10], and other derivative algorithms [11]. However, spectral responses can be easily affected by atmospheric effects, the imaging sensor, vegetation cover, and the presence of soil. Even for the same type of rock, the actual spectral response can be critically different from a pure sample.

To further address the above problems, different types of ML methods have been used to capture discriminative geological features from remote sensing images. The first category of ML methods is dimensionality reduction, including principal component (PC), independent component analysis (ICA) [12], and minimum noise fraction (MNF) [13]. The second category is classification methods, such as maximum likelihood estimation (MLE) and support vector machine (SVM). Yu *et al.* [14] used an SVM with multisource remote sensing data for automated lithological classification. Meanwhile, some clustering methods, such as ISODATA [15] and Kmeans, and regression methods, such as multilinear regression [16], have been utilized to identify lithology, minerals, and alteration types. Kumar *et al.* [17] evaluated the effect of three ML methods, random forest, logistic regression, and SVM on the identification of salt-affected soils, in which the test results of the three ML methods exceeded 95%. ML-based methods can automatically extract features and improve the scalability and compatibility of interpretation models. However, with the continuous improvement of spatial and spectral resolution and data volume, the feature representation of ML-based methods has been challenged.

After 2012, deep learning (DL) methods, which automatically model complex and large-scale data to extract high-level semantic features, have dominated image classification [18], object detection [19], [20], sematic segmentation [21], and weakly supervised learning [22], [23]. Therefore, DL methods are used to address the shortcomings of ML-based methods and have demonstrated their superiority in computer vision. Despite the remarkable performance of DL models, only a few works exist that apply DL models for geological remote sensing interpretation. Latifovic *et al.* [24] used convolutional neural networks (CNNs) to map geological target features and classify geological classes with Landsat images. Sang *et al.* [25] presented a DL framework to extract high-resolution target features for lithological mapping. Shirmard *et al.* [26] used CNNs and classic ML methods to effectively and quickly recognize lithological units for mineral prospectivity mapping. Zhang *et al.* [27] evaluated the effect of DL approaches in automated glacier detection, and extensive experiments proved that DL methods have a promising generalization ability on different remote sensing data. Zhang *et al.* [27] utilized DL technologies and object-based image analysis (OBIA) to build a unified workflow for the detection of rock glaciers, which is an essential element of the cryosphere. Isikdogan *et al.* [28] applied a data-driven DL approach to producing large-scale, high-quality water maps. Generally, DL methods can have broad application prospects in geological remote sensing interpretation, and additional valuable work needs to be carried out.

Regarding the types of remote sensing data used, early researchers used multispectral remote sensing images with low or medium spatial resolutions, including those from the Advanced Spaceborne Thermal Emission and Reflectance Radiometer (ASTER), Landsat Multispectral Scanner (MSS),

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and Sentinel series. With the advancement of imaging technology, high-spatial and hyperspectral images have been acquired and used. The hyperspectral data mainly include images from Hyperion, AVIRIS, and AVIRIS-NG. High-spatial resolution imaging sensors are used in the WorldView series satellites. These high-spatial-resolution and hyperspectral images can capture more fine-grained information from geological remote sensing elements and provide data support for effective and accurate geological remote sensing interpretation. Multiple data sources can also provide the supplementary information to solve the limitations of a single data source. Many scholars have combined multitemporal and multisource data to address the high interclass similarity of geological elements. Wan *et al.* [29] developed an image and point data fusion algorithm to use high-spatial-resolution and time-series images, and LiDAR data for forest stand classification. For deriving water depths, Mandlburger *et al.* [30] introduced a BathyNet, a U-net-like architecture, to process multispectral RGBC (red, green, blue, and coastal blue) and high-resolution aerial images. Kruse and Perry [31] used Worldview 3 and AVIRIS data to explore minerals, in which Worldview 3 images with detailed spatial information were used to map complex alteration patterns of minerals. Also, for mineral mapping, Sekandari *et al.* [32] applied four different data sources to explore the deposits in the metal deposits. With the continuous availability of remote sensing data, multitemporal and multisource images of the same area have become easy to access [33], which will increasingly play a critical role in the interpretation geological elements.

In order to address the problems of numerous classes, high interclass similarities, complex distributions, and sample imbalances of geological elements, we present a novel DL-feature-based adaptive multisource data fusion network (AMSDFNet) for geological remote sensing interpretation. In the proposed AMSDFNet, a series of critical components are used to achieve a satisfying classification performance of multiclass geological elements. It has two branches to learn the features from two different multispectral images independently. In each branch, an ImageNet [34] pretrained feature extractor (ResNet 50 [35] in this model) is first used to simultaneously extract and retain the DL features of the input image. Next, atrous spatial pyramid pooling (ASPP) module [36] is sampled in parallel with the dilated convolution to ensure that AMSDFNet can accurately learn context and boundary information of different types of geological elements. The features extracted from the two branches are then fused and reactivated by a spatial and channel attention block [37], which can adaptively strengthen the weights of the beneficial features and suppress invalid information. To reduce the effect of a severe sample imbalance of geological elements, a hard example mining algorithm [38] is applied to select important training samples and make AMSDFNet focus on learning them.

The experimental area in western China has sufficient geological elements, such as rocks, soil, surface water, and glaciers. Two different data sources used are Landsat 8 and Sentinel 2 images. Extensive experiments were performed to verify the proposed AMSDFNet. The results display that our model improved the two evaluation metrics by more than 2%. This work's contributions are outlined in three aspects, which are as follows.

1) As far as we know, this study is the first to apply DL features and multisource data to interpret multiple geological remote sensing elements of lithology, soil, surface water, and glaciers. The existing DL-based works mainly interpret a single class of geological elements.
2) By focusing on addressing numerous classes, high similarities, complex distributions, and sample imbalances of geological elements, we have developed an efficient AMSDFNet to adaptively fuse valuable information from multisource remote sensing data and mine difficult training samples to boost the performance of geological element segmentation.
3) A region with sufficient geological elements in China was set as the research area. The extensive experiments demonstrated the superiority of the proposed interpretation model.

II. Research Area and Multisource Data

A. Research Area

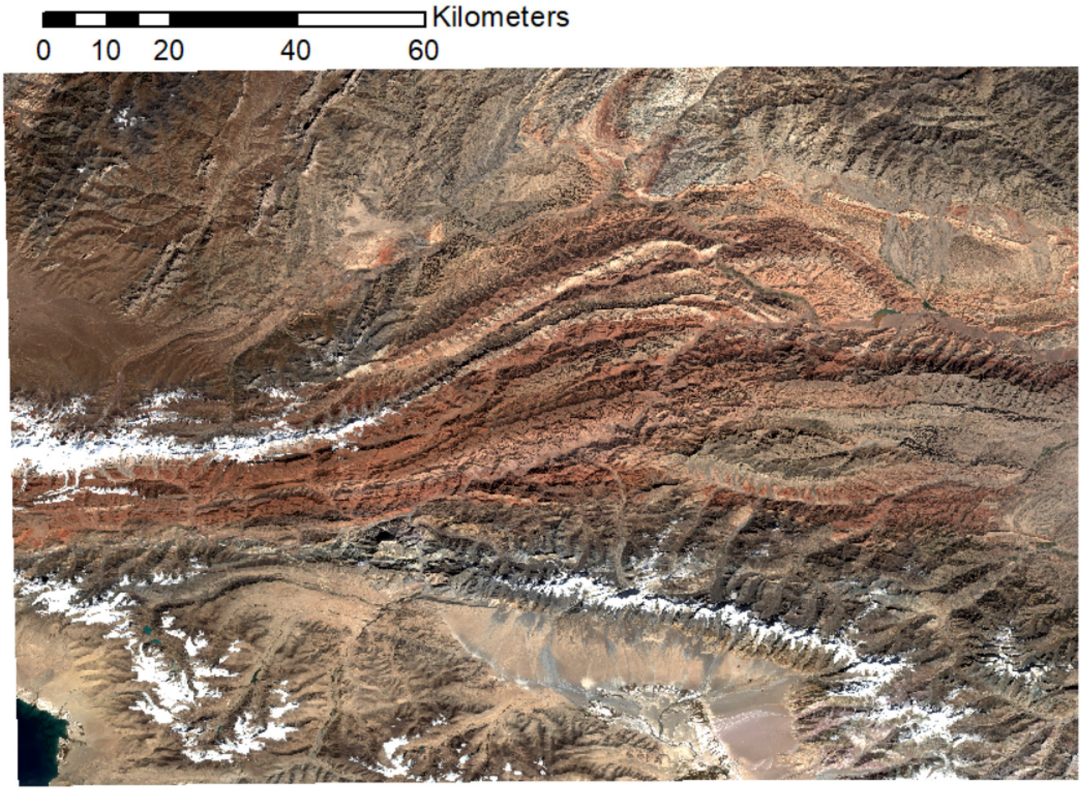
The research area was a part of the intersection of Tarim Basin in western China and the northern foot of the Tianshan and Kunlun Mountains in southwest China. Affected by an arid climate in a temperate zone, the density of outcrops is high. The exposed strata have continuous ages and a wide variety of types, which are suitable for remote sensing satellites to observe lithology and other geological remote sensing elements. Also, influenced by strong tectonic movement and continuous erosion, including the erosion of rivers, the area has formed rich geological elements with a wide variety of lithology, such as sedimentary, magmatic, and metamorphic rocks, in addition to rich glaciers, surface water, and soil. Therefore, it satisfied the needs of the interpretation task of geological elements in this work.

As the main part of regional geological elements, lithology interpretation is the most essential task in the interpretation process. According to the spectral information and spatial characteristics of remote sensing images, determining the lithology and output characteristics of exposed rocks is the main task of the interpretation. Therefore, this study uses multispectral and middle-resolution Sentinel 2 and Landsat 8 images. The We used Sentinel 2 data to extract the spatial and spectral characteristics of images, and Landsat 8 images were used to capture the color characteristics of the geological elements. The two kinds of data sources are described as follows.

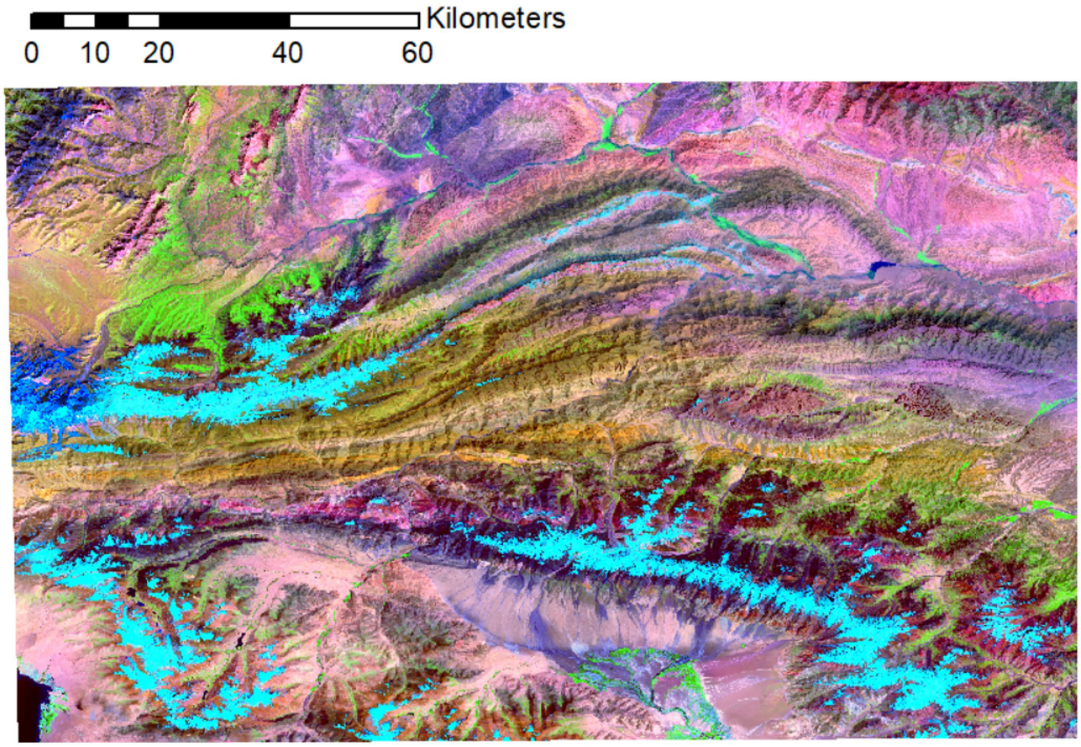
B. Two Kinds of Data Sources

1) Sentinel 2: This is a two polar orbiting imaging satellite with 13 spectral bands. In this study, we used the secondary surface reflectance data of Sentinel 2 that had been archived on the Google Earth Engine (GEE) platform in 2019 and selected bands 2, 3, 4, and 8 with resolutions of 10 m to synthesize an image, in which band 8 was the near-infrared band. The synthesized image is shown in Fig. 2(a).

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(a)



(b)

Fig. 2. Two data sources for the research area. (a) Sentinel 2 image and (b) Landsat 8 image.

2) Landsat 8: This is a multispectral imaging satellite with 11 bands. Band 8 is a panchromatic image with a 15-m spatial resolution. The remaining bands have a 30-m spatial resolution. In this study, we used the secondary surface reflectance

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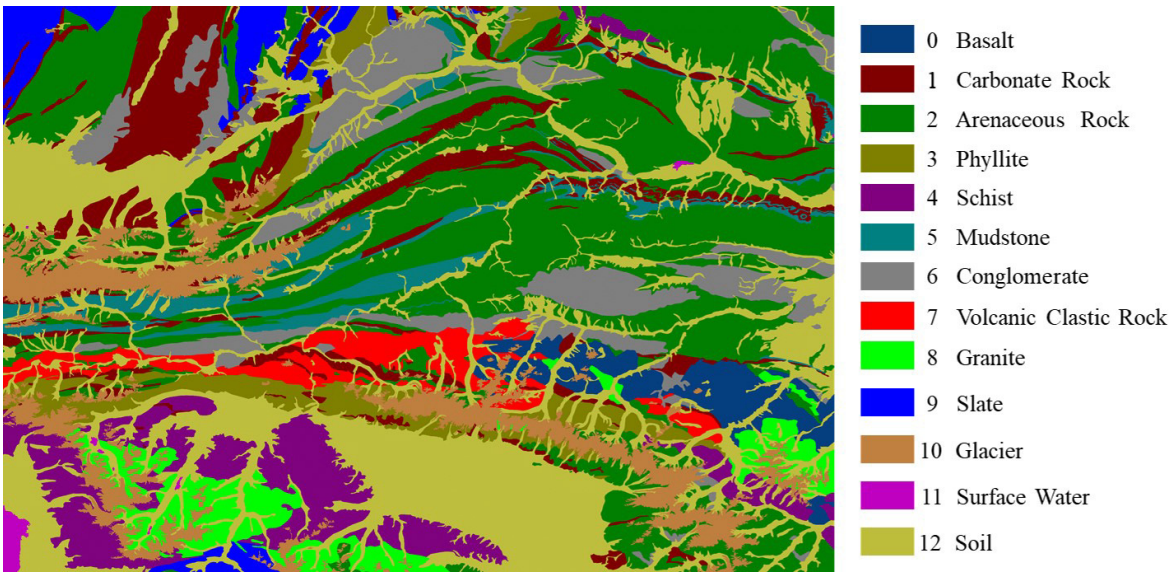


Fig. 3. Annotation map of the research area.

data of Landsat 8 that had been archived on the Google Earth Engine (GEE) platform in 2019. To improve the spatial resolution, we selected the bands of 2, 3, 4, and 8 in the Landsat 8 data to synthesize a new image and then removed the band 8 information. The fusion image's resolution was 15 m. Next, a gamma stretch operation with a coefficient of 1.146 was used in the ArcGIS software to enhance brightness changes in the image, as shown in Fig. 2(b).

C. Data Annotation

The label map from this study is shown in Fig. 3. It was from the geological map of Xinjiang published by the China Geological Survey. Lithology interpretation often regards the class of rock, which involves a combination of lithologies dominated by certain rocks. On the basis of this geological map, the lithology was further divided and combined by geological experts and finally formed into 13 categories, including sedimentary rock, magmatic rock, metamorphic rock, glacier, surface water, and soil. Sedimentary rock includes conglomerate, arenaceous rock, mudstone, and carbonate rock. Magmatic rock includes basalt, granite, and volcanic clastic rock. Metamorphic rock includes slate, schist, and phyllite. These effectively summarized the main geological elements of the area. The width and height of the annotation map were resampled to the size of the Landsat 8 image in Fig. 2.

III. Methodology

This section details the proposed multibranch AMSDFNet, which mainly consists of multisource feature extraction and adaptive data fusion modules. To further address the sample imbalance of geological elements, a hard example mining algorithm was applied to enhance the weights of important training samples and suppress easy samples. Fig. 4 shows the architecture and processing flowchart of the AMSDFNet, which includes the following.

1) *Multisource Feature Extraction:* The AMSDFNet has two branch networks with the same Deeplab v3 [36] architecture, which consists of an ImageNet pretrained CNN and an ASPP module. Two branches can simultaneously extract visual and spectral features from the Sentinel 2 and Landsat 8 images. Next, two features are stacked together via a concatenation operation. Therefore, the channels of the stacked features become doubled, while their width and height have not changed.
2) *Adaptive Feature Fusion:* Due to the large size of the concatenated features, their valuable information may be affected. Therefore, an adaptive feature fusion block calculates the concatenated features' channel and spatial attention weights. Then, the features are rescaled according to the above weights and generate the final segmentation map. This block can strengthen the valuable information and suppress the redundant information in the fused features.
3) *Hard Sample Mining:* In addition, to address the severe problem of sample imbalance in geological elements, we utilize a hard example mining algorithm to automatically select critical samples for better training the proposed AMSDFNet. The algorithm can sort the pixel samples according to their update gradients. The AMSDFNet would learn only pixels with the gradients more than a predefined threshold.

A. Multisource Feature Extraction

Fig. 4 shows each branch that mainly contains an ImageNet pretrained backbone and an ASPP module, which is the main innovation in DeepLab v3 [36] to improve the convolution operation's field-of-view. An additional 3×3 convolution layer was added at the beginning of each branch to uniformly transform the multispectral remote sensing data to 3-D features. The backbone (ResNet 50 in this model) is then used to extract valuable texture, color, and some semantic

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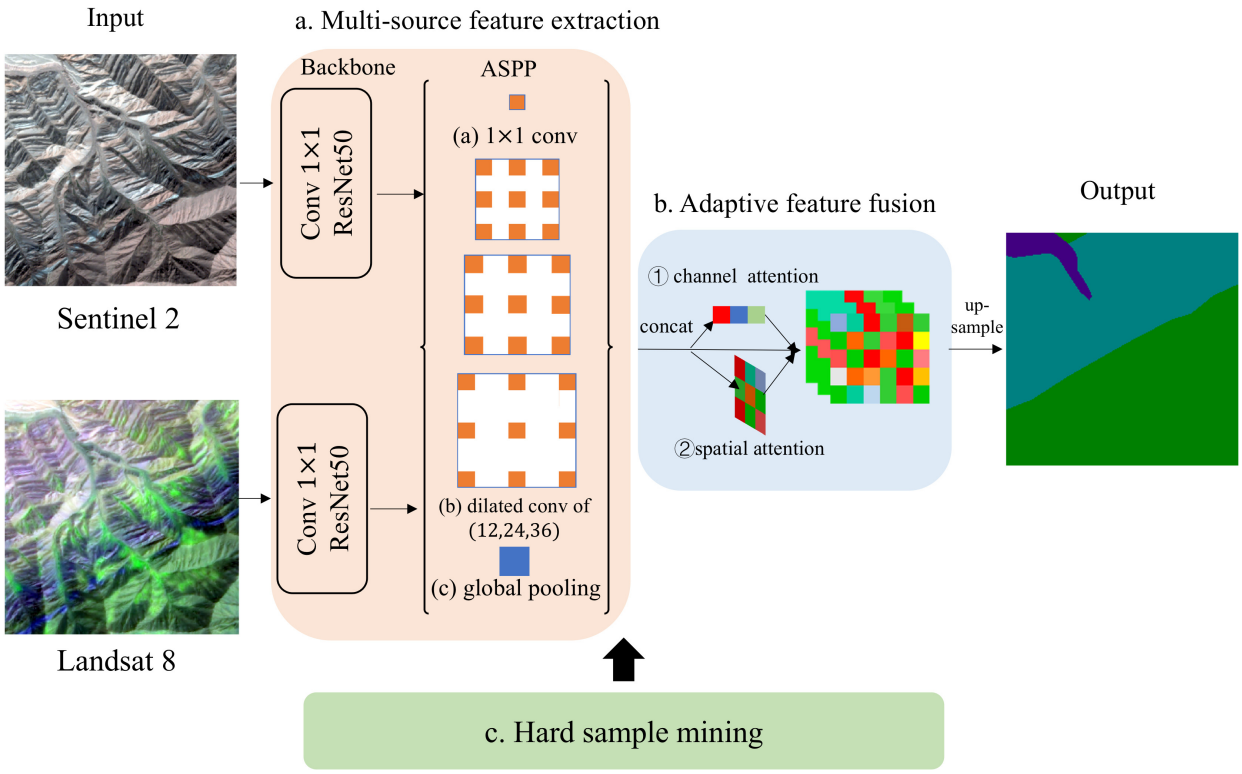


Fig. 4. Overall architecture of the proposed AMSDFNet.

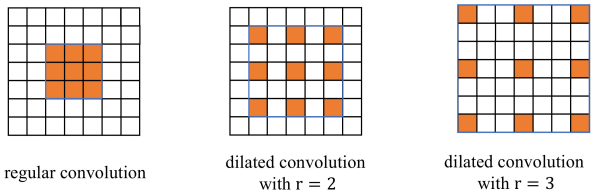


Fig. 5. Comparison of the regular convolution and the dilated convolution.

features from them. Next, the ASPP module [36] can extract multiscale discriminative information according to the output of the backbone.

Multisource remote sensing images provide sufficient multiscale detail and spectral information, which is critical in interpreting geological elements. The ASPP module plays an important role in simultaneously capturing multiscale spatial and spectral features. It consists of one regular 1×1 convolution for extracting valuable channel information, three 3×3 dilated convolutions with dilated rates of $(12, 24, 36)$ for modeling multiscale spatial features, and a global average pooling (GAP) for learning an image-level feature.

As shown in Fig. 5, in contrast with a regular convolution operation, dilated convolution can change its field-of-view by setting different dilated rates r. For an input feature map X, each location i's output Y_i can be computed as follows:

$$Y_i = \sum_k X_{i+r\cdot k} W_k \tag{1}$$

where W are the dilated convolution's weights and k is the index. The dilated rate r is the sampled stride over the input feature. The dilated convolutions with different rates $(12, 24, 36)$ can adjust their field-of-views and extract multiscale, dense spatial feature responses without additional parameters. Therefore, it is conducive to accurately distinguishing the boundaries of different geological elements.

A GAP computes the global pixel average of each feature channel as an image-level feature. For the input features $X \in \mathbb{R}^{C\times H\times W}$, a GAP can generate output features $Y \in \mathbb{R}^{C}$. f is a channel of X, and then, the corresponding output $V_{f,\text{ave}} \in Y$ can be calculated as follows:

$$V_{f,\text{ave}} = \frac{1}{H \times W} \sum_{h=0}^{H-1} \sum_{w=0}^{W-1} f_{h,w}. \tag{2}$$

The ASPP module simultaneously uses a regular 1×1 convolution $\text{Conv}_1(\cdot)$, three different dilated convolutions $\text{dc}_j(\cdot)$ (j is the dilated rate), and a GAP unit to compute the outputs. Next, the output features are resized to the width and height of X by a bilinear interpolation $\text{Bi}(\cdot)$ and then connected in the channel dimension by a concatenation function $\text{Concat}(\cdot)$. Therefore, the calculation process of the ASPP module can be summarized as follows:

$$\begin{aligned} Y = \text{Concat}[&\text{Conv}_1(X);\ Bi(dc_{12}(X));\\ &Bi(dc_{24}(X));\ Bi(dc_{36}(X));\ Bi(\text{GAP}(X))]. \end{aligned} \tag{3}$$

Finally, two branches, the Landsat 8 features Y_{LS} and the Sentinel 2 features Y_{SE}, are stacked together as a whole by the above function $\text{Concat}(\cdot)$.

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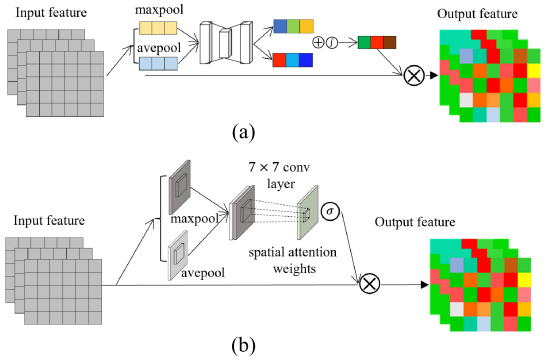


Fig. 6. Schematic of the channel attention and the spatial attention. (a) Channel attention structure. (b) Spatial attention structure.

B. Adaptive Feature Fusion

After the concatenation operation, the valuable information extracted from two data sources is aggregated to interpret geological remote sensing elements. However, the increase of feature dimensions may cause a new problem: while the valuable information is sufficient, some invalid and redundant information also rises rapidly, affecting the final interpretation results. To address this problem, we perform an attention operation in the channel and the spatial dimensions of the concatenated features, which is able to discover the informative regions and channels of the features and improve the interpretation accuracy. The attention weights are calculated by a convolutional block attention module (CBAM) [37], in which a channel attention module (CAM) computes the relationships between the feature channels and a spatial attention module (SAM) discovers the informative areas. The architectures of CAM and SAM are shown in Fig. 6.

1) CAM: As shown in Fig. 6(a), for the input feature map $X \in \mathbb{R}^{C\times H\times W}$, the CAM considers each feature channel as two types of channel descriptors and exploits the features' interchannel relationships. Two types of channel descriptors are generated by jointly using a global max pooling (GMP) and a GAP on the input features. The GAP is computed as (2). For a channel $f \in X$, the GMP would calculate the channel descriptor $V_{f,\max}$ as follows:

$$V_{f,\max} = \max_{h=0,\ldots,H-1} \max_{w=0,\ldots,W-1} f_{h,w}. \tag{4}$$

After the above calculations, two C-channel descriptors are denoted as $D^c_{\max} \in \mathbb{R}^C$ and $D^c_{\text{avg}} \in \mathbb{R}^C$. A two-layer fully connected network is then applied to produce the channel attention maps $M^c_{\max} \in \mathbb{R}^{C\times 1\times 1}$ and $M^c_{\text{ave}} \in \mathbb{R}^{C\times 1\times 1}$. The overall feature descriptor M^c_{all} is computed by element-wise summation of $M^c_{\max}$ and M^c_{ave}

$$M^c_{\text{all}}(X) = \sigma\left(W_{fc}(D_{\max}) + W_{fc}\left(D_{\text{avg}}\right)\right) \tag{5}$$

where W_{fc} is the fully connected network's weights and σ is the sigmoid function.

2) SAM: As shown in Fig. 6(b), this module focuses on discovering the informative regions from the input feature X. Therefore, similar to the CAM, two types of pooling operations are performed in X's channel dimension to generate two 2-D spatial descriptors: $D^s_{\max} \in \mathbb{R}^{H\times W}$ and $D^s_{\text{ave}} \in \mathbb{R}^{H\times W}$, which denote the max-pooling feature and average-pooling feature across X. Next, $D^s_{\max}$ and D^s_{ave} are concatenated and processed by a 7×7 convolution kernel to generate a spatial attention matrix M^s_{all}. M^s_{all} is calculated as follows:

$$M^s_{\text{all}}(X) = \sigma\left(\text{Conv}_7\left(\text{Concat}\left[D^s_{\max}; D^s_{\text{ave}}\right]\right)\right) \tag{6}$$

where Conv_7 is the convolution layer.

Two attention modules are placed in a sequential manner, and the output feature $\tilde{F}$ is rescaled by multiplying M^c_{all} and M^s_{all}

$$\tilde{X} = M^s_{\text{all}}\left(M^c_{\text{all}}(X)\right). \tag{7}$$

C. Hard Example Mining Algorithm

Sample imbalance is a common critical problem in remote sensing. Taking the three main lithologies as examples, sedimentary rocks account for only 7.9% of the crustal mass but occupy up to about 75% of the Earth's land area, while magmatic rocks and metamorphic rocks cover the remaining space. In this case, DL models usually tend to learn geological elements with a large number of samples and ignore elements with sparse distributions, making geological interpretation maps unable to meet the requirements of the related applications. Meanwhile, the samples with sparse distributions usually produce large gradients during the model's training process. Therefore, the gradient values can distinguish whether a pixel is an important sample. We can then make the model focus on learning such samples. The above idea can be implemented as a hard sample mining algorithm, which is described in Algorithm 1.

Algorithm 1 Descriptions of the Hard Example Mining Algorithm

1: **Hyperparameter:** Loss threshold $\tau = 0.7$, minimum number of training samples $N_{min} = 100,000$
2: **Input:** Predict results $pres$, annotation map $labels$
3: **Output:** mean loss value $loss_mean$
4: $CELoss \leftarrow CrossEntropyLoss()$
5: $loss_all \leftarrow CELoss(pres, labels)$
6: $loss_sort \leftarrow Sort(loss_all)$
7: **if** $loss_sort[N_{min}] > \tau$ **then**
8: $\quad loss \leftarrow loss_sort[loss_sort > \tau]$
9: **else**
10: $\quad loss \leftarrow loss_sort[: N_{min}]$
11: **end if**
12: **Return** $mean(loss)$

In Algorithm 1, two hyperparameters are predefined, which are the loss threshold τ and the minimum number of samples $N_{\min}$. Only pixels with a loss value greater than τ will be used to train the AMSDFNet. $N_{\min}$ represents the minimum number of each training batch. For a given predicted segmentation map pres and annotation map labels, a cross-entropy function is initialized to compute the loss values between them (lines 4 and 5 in Algorithm 1). The pixel samples are then sorted according to their loss values (line 6 in Algorithm 1). If there

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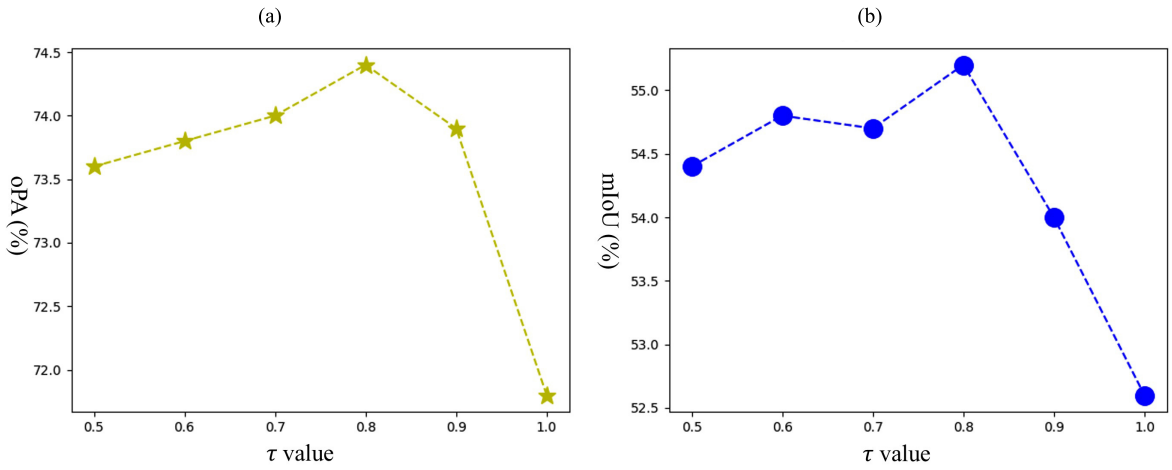


Fig. 7. Experimental results of sensitivity analysis for τ in hard example mining algorithm. (a) oPA value. (b) mIoU value.

are more than $N_{\min}$ pixels with a loss greater than τ, these pixels are used to train the DL model; otherwise, the first $N_{\min}$ samples are selected to participate in the training process (lines 7–10 in Algorithm 1). Finally, the average of the selected samples' loss is set as the overall loss of the DL model.

IV. Experiments and Analysis

In this section, we detail several comparison experiments that evaluated the interpretation performance of our AMSDFNet over a geological remote sensing dataset in Section II. The experimental settings and evaluation metrics are reviewed first. Then, a sensitivity analysis is performed to evaluate the effect of different loss thresholds τ in the hard example mining algorithm on the overall performance. Next, we carry out an ablation study to verify the effectiveness of different modules in the AMSDFNet. Finally, the proposed network is compared with milestone DL semantic segmentation models. The results of pixel accuracy (PA), overall PA (oPA), intersection over union (IoU), mean IoU (mIoU), confusion matrices, predicted segmentation map, and computational complexity are reported.

A. Experimental Settings

The Sentinel 2, Landsat 8, and annotation images introduced in Section II were aligned and segmented into 1101 image patches without overlap. The Landsat 8 and annotation image patches were with 224 × 224 pixels, while the Sentinel 2 image patches were cropped and resized to 448 × 448 pixels to facilitate feature fusion. The total dataset was randomly split into 80% for training and 20% for testing. In this study, some classic DL semantic segmentation models were adopted for comparison: fully convolutional networks (FCNs) [39], dual attention networks (DANets) [40], DeepLab v3 [36], and a lite reduced ASPP (LR-ASPP) network [41]. These backbone was ResNet 50 [35]. We used their pretrained models provided by Pytorch [42] in the experiments. Some of the latest models, including vision transformer (VIT) [43], bilateral network with guided aggregation (BiSeNet V2) [44], and high-resolution network (HRNet) [45], have also been added as the comparison methods, which are implemented by MMSegmentation [46].

The experiments were performed on a workstation equipped with an Intel i7 11700k CPU and an NVIDIA RTX 3090 GPU. The batch size was eight, and the initial learning rate was set to 0.003. The entire training process for each DL model was 120 epochs.

B. Evaluation Metrics

In terms of segmentation accuracy, our experiments adopted two popular metrics [47] to evaluate the overall accuracy.

1) The PA is the simplest metric, which calculates a ratio between the amount of correctly classified pixels and the total pixels. Naturally, the oPA is derived by counting the proportion of all classes' correctly classified pixels.
2) The IoU is a common metric in semantic segmentation. It represents the ratio between the intersection and the union of two pixel sets: our predicted segmentation and the annotation map. The IoU is computed on a single-class sample, and the mIoU is obtained by averaging the IoUs of all the classes.

In terms of model complexity, the number of parameters and the giga floating-point operations per second (GFLOPS) were utilized to measure the size and computational complexity of each DL model.

C. Sensitivity Analysis

In this section, we have evaluated the effect of loss threshold τ in Algorithm 1 on the overall performance of the proposed method. The results are shown in Fig. 7. The values of τ range from 0.5 to 1. Fig. 7(a) shows the change curve of the oPA with different τ's and Fig. 7(b) shows the change curve of the mIoU.

It can be seen that while τ was set to 0.5, the oPA metric of the AMSDFNet was 73.6%, and the mIoU was 54.4%. Then, the overall performance of the AMSDFNet showed an upward trend with an increase of τ. When τ was 0.8, the model performance was the optimum, in which the oPA

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TABLE I
ABLATION STUDY OF THE AMSDFNET

Method	Landsat 8	Sentinel 2	Multi-source	CBAM	Hard example mining	oPA (%)	mIoU(%)
DeepLab v3 (Base model)	√					71.2	52.5
DeepLab v3 (Base model)		√				70.4	49.0
AMSDFNet			√			72.4	53.0
AMSDFNet			√	√		72.8	54.2
AMSDFNet			√	√	√	74.4	55.3

TABLE II
PA AND oPA RESULTS OF THE AMSDFNET AND MILESTONE DL MODELS (%)*

Method Name	Data source	BA	CR	AR	PH	SC	MS	CO	VCR	GR	SL	GL	SW	SO	oPA
FCN [39]	Landsat 8	46.9	57.3	63.0	64.5	69.9	54.3	75.5	**72.0**	67.4	51.1	90.1	95.0	78.3	69.1
FCN	Sentinel 2	40.6	51.2	67.4	59.6	65.0	50.4	75.0	56.2	67.5	39.1	84.6	92.9	79.0	68.6
LRASPP [41]	Landsat 8	45.8	49.7	62.1	60.6	68.7	53.6	65.7	60.5	64.6	59.8	90.4	99.5	78.3	66.7
LRASPP	Sentinel 2	53.5	49.6	68.9	57.5	73.5	48.7	68.6	71.2	60.4	54.1	87.9	**99.9**	82.9	70.0
DANet [40]	Landsat 8	58.0	61.4	66.3	69.7	68.6	**66.2**	74.3	60.0	69.1	68.0	89.9	98.6	81.1	71.8
DANet	Sentinel 2	47.2	53.2	68.1	57.9	61.8	41.9	74.2	55.5	64.9	52.8	86.7	97.9	79.9	69.4
VIT [43]	Landsat 8	58.9	**64.6**	77.4	**72.7**	71.9	59.4	65.1	62.9	48.9	67.5	95.9	59.5	75.2	72.5
VIT	Sentinel 2	40.0	37.3	68.7	26.1	39.8	20.4	29.1	30.7	22.0	31.1	83.4	89.3	65.5	52.9
Bisenetv2 [44]	Landsat 8	**77.6**	59.5	77.4	67.6	50.5	17.7	50.6	33.3	67.9	**72.4**	96.3	91.3	86.2	70.6
Bisenetv2	Sentinel 2	54.2	58.1	**77.7**	40.0	73.2	25.6	40.6	21.6	18.7	41.2	91.1	86.9	80.3	65.3
HRNet [45]	Landsat 8	71.4	61.4	75.5	47.1	69.3	40.3	45.5	58.1	53.5	50.0	**96.7**	98.8	**88.4**	70.8
HRNet	Sentinel 2	72.2	49.7	**77.7**	33.0	68.3	40.8	38.7	56.5	35.9	41.7	92.3	90.6	85.7	67.4
DeepLabv3 [36]	Landsat 8	50.9	57.3	66.3	63.1	76.0	56.3	71.5	63.9	**73.8**	64.3	90.9	86.1	81.4	71.2
DeepLabv3	Sentinel 2	49.3	55.1	66.8	61.0	70.3	62.7	79.6	62.6	65.3	34.2	86.9	91.5	81.2	70.4
Our Method	multi-source	54.8	63.1	69.1	71.0	**76.8**	61.7	**82.6**	70.0	72.4	36.3	90.5	95.2	85.4	**74.4**

* Basalt (BA), Carbonate Rock (CR), Arenaceous Rock (AR), Phyllite (PH), Schist (SC), Mudstone (MS), Conglomerate (CO), Volcanic Clastic Rock (VCR), Granite (GR), Slate (SL), Glacier (GL), Surface Water (SW) and Soil (SO).

was 74.4% and the mIoU was 55.3%. Subsequently, as τ continued to increase, the performance of the AMSDFNet dropped rapidly. The possible reason was that a large τ would filter out a large number of samples that should be learned, resulting in a poor learning degree of the model for the training samples. Therefore, we set $\tau = 0.8$ to carry out the remaining experiments.

D. Ablation Study

In this experiment, an ablation study has been carried out to evaluate the effectiveness of different components in the AMSDFNet. Because our method is implemented based on DeepLab v3, it was set as the baseline model. The experimental results are collected in Table I.

In Table I, the baseline model with Landsat 8 images achieved an oPA of 71.2% and an mIoU of 52.5%, which outperformed the results with Sentinel 2 images by about 0.8% oPA and 3.5% mIoU. One possible reason is that the rich tonal information of Landsat 8 images played a crucial role in interpreting geological elements. By fusing two data sources, the AMSDFNet reached an oPA of 72.4% and an mIoU of 53.0%, which was better than the results with the single-source data. The results proved that multisource remote sensing images could provide more abundant information than single-source images. After adding the CBAM module, our method improved the performance by about 0.4% oPA and 1.2% mIoU. Based on the multisource data fusion and the CBAM, the hard example mining further boosted the overall performance, which reached an oPA of 74.4% and an mIoU of 55.3%. The above experimental results demonstrate the positive effects of the multisource data, the CBAM, and the hard example mining in the AMSDFNet.

E. Comparison Experiments With Milestone Models

These experiments evaluated the interpretation performance of the AMSDFNet with several representative DL semantic segmentation models. The existing DL models were designed to process single-source images. Therefore, the DL compar-

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TABLE III
IoU AND mIoU RESULTS OF THE AMSDFNET AND MILESTONE DL MODELS (%)*

Method Name	Data source	BA	CR	AR	PH	SC	MS	CO	VCR	GR	SL	GL	SW	SO	mIoU
FCN	Landsat 8	42.1	40.0	54.3	37.4	50.4	21.4	38.4	42.0	38.0	33.8	87.0	93.9	69.3	49.8
FCN	Sentinel 2	27.6	37.6	55.9	36.9	47.8	25.0	43.4	29.5	34.1	26.5	77.2	91.3	70.1	46.4
LRASPP	Landsat 8	41.5	31.7	50.8	30.1	53.2	24.5	37.9	42.2	34.0	**48.8**	86.6	80.8	65.9	48.3
LRASPP	Sentinel 2	50.6	39.2	56.7	24.3	54.1	28.6	44.7	43.4	44.5	29.8	78.2	54.1	70.9	47.6
DANet	Landsat 8	48.0	45.1	57.2	42.9	54.9	26.8	44.9	35.6	36.8	46.7	86.4	83.4	71.3	52.3
DANet	Sentinel 2	40.0	37.4	56.8	32.7	49.6	19.1	45.6	32.0	33.2	31.7	79.2	91.7	70.9	47.4
VIT	Landsat 8	41.3	**48.0**	**61.0**	41.0	56.3	**45.1**	**53.7**	46.9	42.0	32.1	**89.4**	56.6	63.2	52.1
VIT	Sentinel 2	23.5	24.9	43.6	19.5	26.3	14.1	21.1	20.2	15.8	21.0	69.3	87.2	44.7	33.2
Bisenetv2	Landsat 8	**51.7**	42.2	55.0	48.9	45.4	14.7	40.4	30.4	**48.4**	46.7	88.9	78.2	71.5	50.9
Bisenetv2	Sentinel 2	38.0	38.6	53.6	27.2	48.1	20.9	32.7	16.4	16.7	31.3	76.1	83.7	64.2	42.1
HRNet	Landsat 8	43.6	45.2	54.8	37.2	53.1	30.5	39.5	41.6	41.9	30.9	88.3	93.6	72.1	51.7
HRNet	Sentinel 2	42.8	36.0	54.1	24.0	49.9	29.0	33.6	38.4	26.5	25.7	76.9	88.2	71.4	45.9
DeepLabv3	Landsat 8	44.6	42.4	55.9	**49.4**	56.7	21.9	42.4	43.1	43.0	41.5	87.2	83.8	70.5	52.5
DeepLabv3	Sentinel 2	45.9	42.4	55.9	25.2	50.9	23.7	50.9	41.9	33.1	25.6	79.6	89.8	71.8	49.0
Our Method	multi-source	51.4	46.4	60.7	45.2	**59.6**	24.1	51.4	**50.2**	42.7	28.5	87.7	**94.6**	**75.8**	**55.3**

* Basalt (BA), Carbonate Rock (CR), Arenaceous Rock (AR), Phyllite (PH), Schist (SC), Mudstone (MS), Conglomerate (CO), Volcanic Clastic Rock (VCR), Granite (GR), Slate (SL), Glacier (GL), Surface Water (SW) and Soil (SO).

ison methods were trained and tested on the Sentinel 2 and Landsat 8 data, respectively. For multispectral data with more than three channels, a preprocessing convolution layer was added at the beginning of the DL models, which was trainable and captured valuable information of the multispectral data into the three channels. The PA per geological class, oPA, IoU per class, and mIoU for all the segmentation models are collected in Tables II and III. The classification confusion matrices and some predicted segmentation maps from DeepLab v3 and our method with multisource data are shown in Figs. 8 and 9.

The interpretation difficulty of different types of geological elements is greatly different. In Tables II and III, it can be seen that the interpretation results of glacier (GL), surface water (SW), and soil (SO) for all of the DL models were relatively better than other geological element categories as most of them had PAs greater than 80% and IoUs greater than 70%. In contrast, the results of phyllite (PH) and mudstone (MS) were terrible. This is because the visual characteristics of soil, surface water, and glacier in remote sensing are quite different, and the DL models can easily distinguish those differences. However, the differences between various kinds of lithologies are small, showing high interclass similarity, which is challenging for the interpretation models to classify them. In addition, the overall accuracy of the DL models with the Sentinel 2 data was worse than that of Landsat 8 data, which is the same as the ablation study.

In terms of the comparison of different DL models, FCN achieved an oPA of 68.6% and an mIoU of 46.4% on Sentinel 2 data, and an oPA of 69.1% and an mIoU of 49.8% on Landsat 8 data. The experimental results of FCN were satisfactory and illustrated the great potential of DL features in the interpretation of geological remote sensing elements. The LRASPP is memory and computationally efficient, and it achieved a similar performance to FCN. Meanwhile, DeepLab v3 and DANet are larger and more advanced models. For the Landsat 8 data, DANet reached a 71.8% oPA and 52.3% mIoU, and DeepLab v3 achieved a 71.2% oPA and 52.5% mIoU, reaching the excellent performance of all the milestone methods compared. Both the ASPP in Deeplab v3 and the dual attention in DANet are good at establishing global contextual and multiscale information, which is significant to interpret multiclass complex geological elements. Some of the latest methods, such as VIT and HRNet, did not show their superiority in the geological element interpretation. Their overall performance is still close to DeepLab v3 and DANet. It is worth noting that VIT's segmentation accuracy on Sentinel 2 was obviously reduced, only reaching a 52.9% oPA and 33.2% mIoU. The possible reason is that the severe problems in geological elements, such as numerous classes, high interclass similarity, and blurred boundaries, are exceedingly different from natural images. These advanced transformer models are more available for natural scenes and have higher requirements for training data and settings. Therefore, it is difficult to apply them to directly process multispectral remote sensing images and interpret geological elements. The proposed AMSDFNet reached 74.4% oPA and 55.3% mIoU, which exceeded the existing methods by about 1.9% and 2.8%, respectively. It also achieved the highest performance in multiple geological element categories. The experimental results have demonstrated the high efficiency of multisource data for geological remote sensing interpretation.

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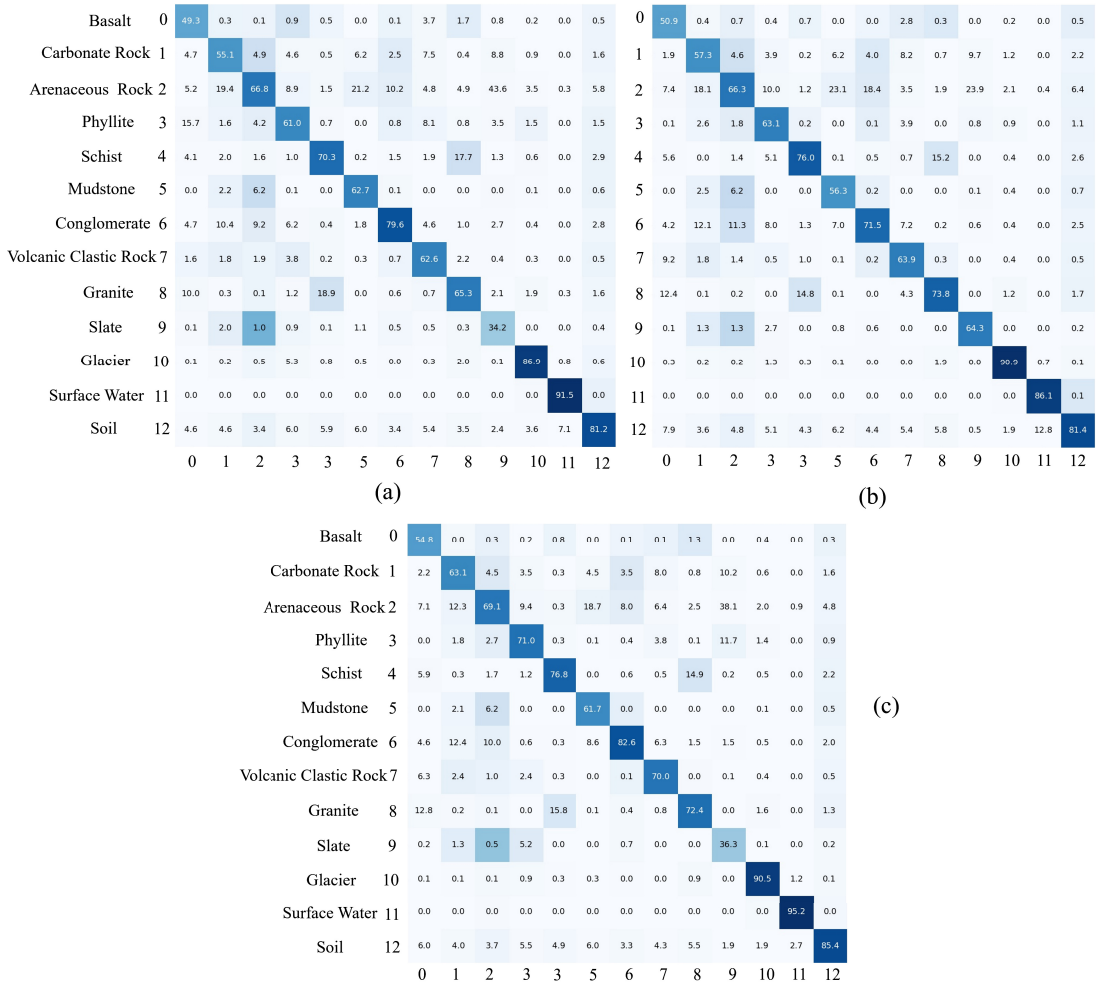


Fig. 8. Confusion matrices of the DeepLab v3 and our AMSDFNet. (a) DeepLab v3 with Sentinel 2. (b) DeepLab v3 with Landsat 8. (c) Proposed method.

Because each branch of our proposed approach was implemented using the ASPP module in DeepLab v3, we mainly compared the classification performance of our model with it. As shown in Fig. 8, our AMSDFNet improved the classification performance of 9 out 13 classes. This shows that valuable information in the two sources of data was retained in the process of the multisource data fusion. Fig. 9 includes some predicted maps of DeepLab v3 with the two different data sources and our segmentation model with multisource data. In Fig. 9(a), our model correctly identified the mixed area of slate, arenaceous rock, and soil, while DeepLab v3 could not classify all the element categories and their boundaries. In Fig. 9(b), the boundary between arenaceous rock and mudstone was the most important part of this image, and our method had satisfactory recognition results. In the red rectangular regions of Fig. 9(c) and (d), the predicted maps of our method were closer to the label than the contrast method.

F. Complexity Analysis

In this experiment, two metrics, the number of parameters and GFLOPs of DL models, were used to evaluate the model complexity. The comparison between the AMSDFNet and Deeplab v3 was carried out and the experimental results are shown in Table IV.

TABLE IV
AMOUNT OF PARAMETERS AND COMPUTATIONAL COMPLEXITY OF THE AMSDFNET AND MILESTONE DL MODELS

Method Name	Input size	Parameters (Milion)	Computational complexities (GFLOPS)
Deeplabv3	224×224×3	42.01	66.52
Deeplabv3	448×448×4	42.01	266.38
Our Method	224×224×3& 448×448×4	84.01	332.74

As can be seen from Table IV, due to the size differences between the Sentinel 2 and Landsat 8 images, the computational overhead of processing Sentinel 2 images for Deeplab v3 is about four times that of processing Landsat 8 images. The proposed AMSDFNet's computational overhead was approximately equal to the sum of the computational costs of Deeplabv3 to process these two kinds of remote sensing images. The channel and spatial attention blocks and the hard example mining algorithm hardly increased the model parameters and computation cost.

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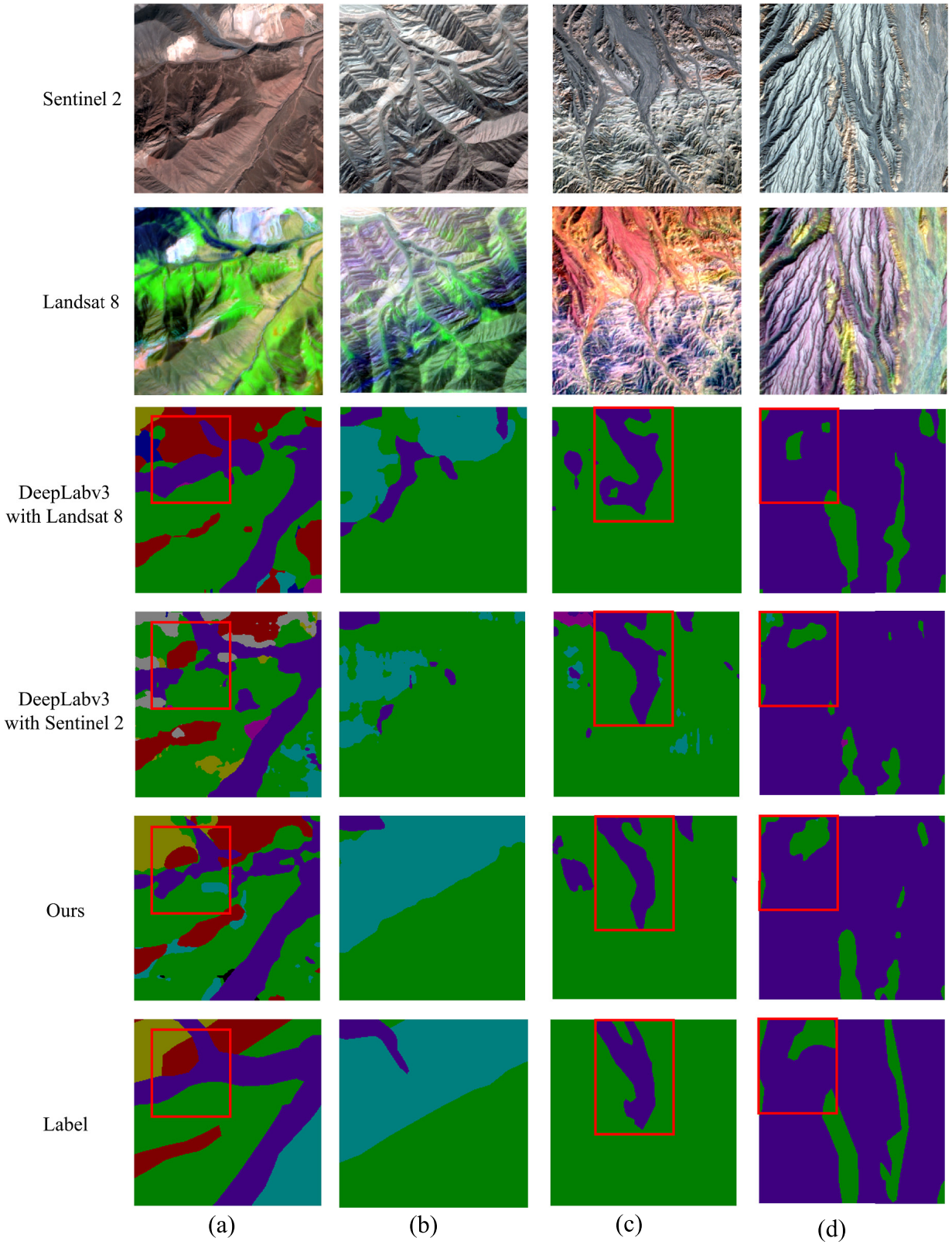


Fig. 9. (a)–(d) columns are four predicted examples for our method and DeepLab v3. The first two rows are the input Sentinel 2 and Landsat 8 images, the middle two rows are the prediction results of Deeplabv3 with two different data sources, and the last two columns are the prediction results of our method and the annotation maps, respectively.

V. Conclusion

In this work, focusing on the problems of numerous classes, high interclass similarities, and sample imbalance of geological remote sensing elements, we presented a multibranch AMSDFNet to combine the advantages of multisource data, feature adaptive fusion, and a hard example mining algorithm

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for the efficient interpretation of geological remote sensing elements. Extensive experiments were performed to verify the proposed model. The study area was a region in western China with sufficient rock, soil, surface water, and glacier. The two kinds of data sources were from Sentinel 2 and Landsat 8 satellites. The experiments showed that our model can improve the oPA and mIoU by about 1.9% and 2.8%, respectively. In the future, we plan to design more efficient multisource fusion algorithms and apply high-resolution and hyperspectral data in geological remote sensing interpretation.

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Wei Han received the B.S. and Ph.D. degrees from the China University of Geosciences, Wuhan, China, in 2015 and 2021, respectively.

He is currently an Associate Professor with the School of Computer Science, China University of Geosciences. His research interests include high-resolution remote sensing image processing, geological remote sensing, and high-performance computing.



Jun Li (Fellow, IEEE) received the B.S. degree in geographic information systems from Hunan Normal University, Changsha, China, in 2004, the M.E. degree in remote sensing from Peking University, Beijing, China, in 2007, and the Ph.D. degree in electrical engineering from the Instituto de Telecomunicações, Instituto Superior Técnico (IST), Universidade Técnica de Lisboa, Lisbon, Portugal, in 2011.

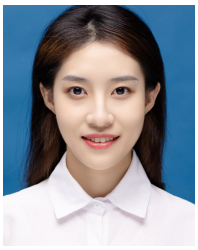
She is currently a Full Professor with the School of Computer Science, China University of Geosciences, Wuhan, China. Her main research interests comprise remotely sensed hyperspectral image analysis, signal processing, supervised/semisupervised learning, and active learning.

Prof. Li is the Editor-in-Chief of the IEEE JOURNAL OF SELECTED TOPICS IN APPLIED EARTH OBSERVATIONS AND REMOTE SENSING. She has been a Guest Editor for several journals, including PROCEEDINGS OF IEEE and *ISPRS Journal of Photogrammetry and Remote Sensing*.



Sheng Wang received the B.S. degree in spatial information and digital technology from the China University of Geosciences, Wuhan, China, in 2021, where he is currently pursuing the M.S. degree in computer science and technology.

His research interests include high-resolution remote sensing image processing, geologic remote sensing, and geoscience knowledge graph.



Xinyu Zhang received the B.S. degree in remote sensing science and technology from the China University of Geosciences, Wuhan, China, in 2021, where she is currently pursuing the M.S. degree in geo-information engineering.

Her research interests include geologic remote sensing, deep learning, and high-resolution remote sensing understanding.



Yusen Dong received the B.S. degree in geology, the M.S. degree in geography, and the Ph.D. degree in cartography and geographic information systems from the China University of Geosciences, Wuhan, China, in 1999, 2002, and 2007, respectively.

From 2006 to 2008, he was engaged in cooperative research in InSAR, DInSAR, and PSInSAR with the University of New South Wales, Sydney, NSW, Australia. He is currently an Associate Professor with the School of Computer, China University of Geosciences. His research interests include DInSAR and ground deformation monitoring, geological remote sensing and remote sensing investigation of land and resources, and glacier change monitoring.



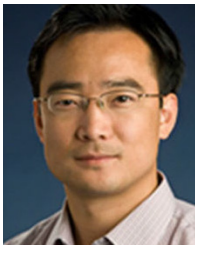
Runyu Fan received the B.S. degree in computer science and technology from the China University of Geosciences, Wuhan, China, in 2017, where he is currently pursuing the Ph.D. degree in geoscience information engineering.

His research interests include urban settlement mapping, land surface intelligent interpretation, and remote sensing data analysis.



Xiaohan Zhang received the B.S. degree in remote sensing science and technology from the China University of Geosciences, Wuhan, China, in 2019, where he is currently pursuing the Ph.D. degree in geoscience information engineering.

His research interests include machine learning, deep learning, unsupervised representation, and high-resolution remote sensing understanding.



Lizhe Wang (Fellow, IEEE) received the B.E. and M.E. degrees in electrical engineering and automation from Tsinghua University, Beijing, China, in 1998 and 2001, respectively, and the D.E. degree in applied computer science from the University of Karlsruhe, Karlsruhe, Germany, in 2007.

He is currently the Dean of the School of Computer Science, China University of Geosciences, Wuhan, China. His research interests include remote sensing data processing, digital earth, and big data computing.

Mr. Wang is a fellow of the Institution of Engineering and Technology (IET) and British Computer Society (BCS) and an Associate Editor of *Remote Sensing* and *International Journal of Digital Earth*.

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